1 Solution — GIAM 2.2 Problem 6

1.1 Problem

Why is it that the sentence “If pigs can fly, I am the king of Mesopotamia.” is true?

1.2 Answer

$$\boxed{\text{Its hypothesis (”pigs can fly”) is } \textbf{false}, \text{ so the conditional is } \textit{vacuously true.}}$$

A conditional $P \Rightarrow Q$ is false in exactly **one** situation — P true and Q false. Here $P =$ “pigs can fly” is false, so that situation cannot occur, and the sentence is automatically true *regardless of whether the speaker is king of Mesopotamia.*

1.3 The Explanation

Let $P =$ “pigs can fly” and $Q =$ “I am the king of Mesopotamia.” The sentence is $P \Rightarrow Q$. Recall the conditional’s truth table:

P	Q	$P \Rightarrow Q$
T	T	T
T	F	**F**
F	T	**T**
F	F	**T**

Table 1.1

The only way for $P \Rightarrow Q$ to be **false** is the second row: hypothesis true, conclusion false. Since “pigs can fly” is false, we are in one of the two bottom rows — and **both give true**. So the sentence is true.

A conditional is false in only one case. With a false hypothesis, that case can never arise.

P = "pigs can fly" — **FALSE**
Q = "I am the king of Mesopotamia" — truth value irrelevant

P	**Q**	**P ⟹ Q**
T	T	T
T	F	**F ← the only false row**
F	T	**T**
F	F	**T**

Both shaded rows (P false) give TRUE — and our sentence lives in a shaded row, since "pigs can fly" is false.

False hypothesis ⟹ the conditional is vacuously TRUE,
no matter what the conclusion says.

Figure 1.1: The setup: P = 'pigs can fly' is FALSE; Q = 'I am the king of Mesopotamia' has irrelevant truth value. The truth table for P implies Q: row T,T gives T; row T,F gives F, the only false row; rows F,T and F,F (shaded) both give T. Our sentence lives in a shaded row because 'pigs can fly' is false, so the conditional is vacuously true no matter what the conclusion says.

1.4 Why the $\neg P \vee Q$ Form Makes It Obvious

Using Problem 2's equivalence $P \Rightarrow Q \equiv \neg P \vee Q$, the sentence becomes

$$\underbrace{\neg(\text{pigs can fly})}_{\text{TRUE}} \;\vee\; (\text{I am king}).$$

Since "pigs *cannot* fly" is true, the disjunction is already true on its first half — the second disjunct never even needs to be examined. This is the cleanest way to see vacuous truth: a false hypothesis makes $\neg P$ true, which satisfies the "or" by itself.

1.5 Remark — The Conditional Makes No Claim When the Hypothesis Fails

The intuition is a *promise*. "If P, then Q" is a pledge that only comes due when P actually happens. If P never happens, the pledge is never called upon, so it cannot be broken — and an unbroken promise counts as kept (true). The speaker who says "if pigs can fly, I'm king of Mesopotamia" risks nothing, because pigs don't fly; there is no possible world (among the actual facts) in which they are caught out. Crucially, the sentence asserts **nothing** about the speaker's royalty — it would be *equally true* with any conclusion: "if pigs can fly, then $2 + 2 = 5$" is just as true.

1.6 Remark — This Is a Rhetorical Device

The construction is used in ordinary speech precisely *because* everyone knows the hypothesis is absurd. By attaching an outlandish conclusion to an impossible condition, the speaker vividly asserts "P is false" — "I'm no more the king of Mesopotamia than pigs can fly." Sarcasm like "if he passed that exam, I'll eat my hat" works the same way: the true content is the *falsity of the hypothesis*, conveyed by daring the listener to accept a ridiculous consequence. Logic's vacuous-truth convention is exactly what makes the rhetorical move valid — the whole conditional is true, so uttering it commits the speaker to nothing embarrassing.

1.7 Remark — Vacuous Truth Is Indispensable in Mathematics

Far from a mere curiosity, vacuous truth is *essential* to how mathematics is stated. Countless true theorems are vacuously true in edge cases:

- "Every element of the empty set is purple" is **true** — there is no element to serve as a counterexample.
- "For all x, if $x^2 < 0$ then $x = 5$" is **true** over the reals, since $x^2 < 0$ never holds.
- "If n is both even and odd, then $n = 100$" is **true**, vacuously.

The convention is forced on us: a universal statement $\forall x\,(P(x) \Rightarrow Q(x))$ should be false *only* if some x actually violates it (satisfies P but not Q). With no such witness, it must be true. Defining the conditional any other way would make basic laws — like "the empty set is a subset of every set" — break down. So the pigs-can-fly sentence is a jokey instance of a convention that quietly underpins theorems throughout the book.

1.8 Remark — How to *Refute* Such a Sentence (You Can't, Here)

To disprove a conditional you must exhibit the one forbidden combination, $P \wedge \neg Q$ (Problem 4) — the hypothesis holding while the conclusion fails. For "if pigs can fly, I'm king," a refutation would require producing a *flying pig* alongside a *non-royal speaker*. Since no flying pig exists, no such counterexample can be assembled, and the sentence stands unrefuted — which is precisely what "true" means for it. This is the mirror image of Problem 4: there, one counterexample kills a conditional; here, the impossibility of even *starting* a counterexample (no true hypothesis) guarantees the conditional survives.